

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

Toni Guan^{a,*}

^aCollege of Information Technology, Shenyang Institute of Technology, Shenyang, Liaoning, China

Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10,000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (+0.0159 vs.

*Corresponding author.

+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, although a source-fitted, label-free temperature predicts the *sign* of the benefit in 60/60 cells); and a discrimination-aware gate (C5) shows 7/12 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [3], Chapman-Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [25]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target

the source-fit temperature is mismatched to the target logits and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [26] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from decay to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [27, 28]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the

premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. 47/60 = 78.3% of ID point estimates are positive and 2/60 = 3.3% have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case, -0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$) recovers ground truth in the identifiable regime (27/27 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime 15/30); after exclusion the EM OOD finding attenuates. The mechanism is prior-likelihood mismatch: EM fits a target prior that amplifies shift-induced miscalibration when the source confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 12 pair $\times$ architecture strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, $\text{acc} = 0.657 < 0.713$). The

gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [8] proposed the static calibration error, and Kumar et al. [18] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [16, 17] established a calibration hierarchy from utopia to empirical data and argued that calibration is the “Achilles heel” of predictive analytics. Strodtthoff et al. [19] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

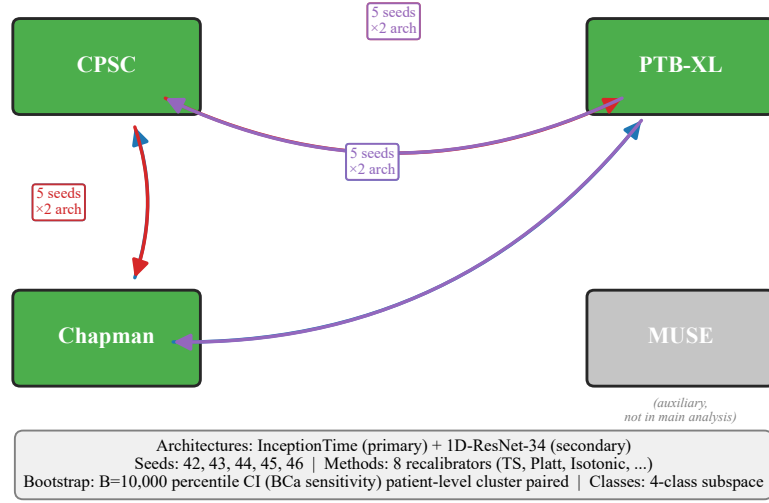


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

120 2.2. *Prior-Shift Estimation and Transfer Calibration*

121 Lipton et al. [11] introduced black-box shift estimation (BBSE) for detect-
122 ing and correcting label shift using confusion-matrix inverses. Alexandari et al.
123 [10] showed that maximum-likelihood calibration with bias correction is hard-to-
124 beat at label shift adaptation, providing the theoretical foundation for our EM
125 prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates
126 transferability for auto-selecting target-domain unsupervised optimal T ; Pseudo-
127 Cal [23] uses pseudo-labels for calibrated transfer; and LaSCal [24] provides label-
128 aware shift calibration with distribution-free guarantees. Our S1 paradigm differs
129 fundamentally: we fit on the *source* calibration split only (no target labels), with
130 S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly
131 weaker assumption than TransCal/PseudoCal, which access target logits or labels.

132 2.3. *Cross-Domain ECG Calibration*

133 Barandas et al. [12] evaluated uncertainty quantification methods in multi-
134 label ECG classification, finding that ensemble-based methods outperform single-
135 model approaches under subpopulation shift. Vranken et al. [13] studied uncer-
136 tainty estimation for 12-lead ECG deep learning and reported significant calibra-
137 tion drift across acquisition devices. Haekal et al. [20] performed cross-dataset val-
138 idation for atrial fibrillation detection with calibration-aware probability analysis,
139 and Patel & Beedala [21] documented calibration drift under cross-institutional
140 ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general
141 vision [1]; we contribute the multi-seed ECG quantification and the boundary-
142 condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

143 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

144 Recent work has introduced ECG foundation models trained on millions of
145 recordings. ECGFounder [29, 37] (10 million ECGs, Harvard–Emory) reaches
146 expert-level performance on 80 diagnoses and transfers cross-domain via fine-
147 tuning. ECG-FM [30] is an open-weight transformer pretrained on 1.4 million
148 ECG segments with hybrid contrastive and generative self-supervision, showing
149 strong cross-dataset generalizability and label efficiency. HeartLang [31] treats
150 heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked
151 sentence pre-training. Tan et al. [32] found that fine-tuning benefits diminish
152 as the downstream dataset grows and that CNNs transfer more effectively than
153 RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD
154 [33] quantifies distribution-shift severity via uncertainty estimation to guide se-
155 lective LoRA layer unfreezing, and BeatRhythm-TTA [34] performs test-time
156 adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al.
157 [35] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-
158 preserving beat-wise augmentation for cross-source generalization, and Chen et al.
159 [36] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned
160 contrastive alignment. These works advance ECG transfer but not the *calibra-*
161 *tion* of transferred models, the gap our study fills via pre-registered cross-corpus
162 evaluation of recalibration methods.

163 3. Data and Pre-registered Protocol

164 Figure 1 summarizes the experimental design.

165 3.1. Datasets

- 166 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the down-
167 loaded `ptbx1_database.csv` copy contains 21,799 (a 38-record deficit in
168 the source CSV, disclosed as a provenance limitation), official 10-fold (folds
169 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records
170 are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- 171 • **Chapman-Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error
172 removed and 21 non-finite records quarantined (`qc_report.json`), for
173 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- 174 • **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records;
175 the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453
176 records; the CPSC2018 recordings not used by the Challenge), both re-
177 leased through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union:
178 CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnos-
179 tic labels. Local extraction retains 6,844 of the 6,877 CPSC Database
180 records (33 records, A6845–A6877, were lost during archive extraction;
181 disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records.
182 Patient-wise split. Two labeling facts must be stated explicitly. First,
183 CPSC2018’s official taxonomy is nine disease classes (Normal, AF, I-AVB,
184 LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the
185 NORM/MI/STTC/CD superclass system we adopt is PTB-XL’s [3], ap-
186 plied here as a cross-corpus harmonization choice, not a CPSC-defined
187 one. Second, CPSC2018’s nine classes contain *no* MI; the MI class in our
188 4-class subspace is populated entirely by CPSC-Extra’s SNOMED anno-

189 tations (164867002, 164865005, 54329005). A strict official-class-name
190 caliber would therefore reduce the subspace to {NORM, CD, STTC}; we
191 retain MI because it is the only way to keep a myocardial-infarction category
192 on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) →
193 main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides
194 of the pair recomputed symmetrically.

195 3.2. *Label Mapping and Leakage Guard*

196 Single-label priority rules are fixed (MI > STTC > CD > HYP >
197 NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements
198 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
199 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
200 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard
201 assertion in the reproducibility repo). Normalization statistics come only from
202 the train split; augmentation applies only to train.

203 4. **Methods**

204 4.1. *Transfer pairs and shift matrix*

205 We study six directed cross-corpus transfer pairs: Chapman→CPSC,
206 Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman,
207 and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per
208 pair before any endpoint is computed, as follows. A pair whose source or target
209 is CPSC2018 is restricted to the 4-class subspace {NORM, CD, STTC, MI}
210 (CPSC’s HYP class has $n = 11$, §3), with both sides of the pair recomputed sym-
211 metrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,

212 and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing
213 and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary
214 {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman
215 ($K = 5$). Each experiment records its own K and label mapping (num_classes,
216 label_map in transfer_result.json), so no endpoint mixes class counts
217 within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or
218 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values
219 across cells; the per-pair tables (Tables 1–4) are therefore the appropriate level
220 at which to read K -specific results. The L2 eval-time shift matrix (no retraining)
221 covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise
222 injection (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5 / \times 2$.

223 4.2. Architecture and seeds

224 We use two architectures: InceptionTime (primary, 46,000–184,000 param-
225 eters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-
226 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The Bi-
227 Mamba architecture is included only as a toy experiment ($n = 60$ records, down-
228 sampled resolution) and deferred to the supplement; the full-resolution BiMamba
229 run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the
230 main-endpoint analysis or the architecture-robustness claim. Training: AdamW +
231 cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training;
232 temperature applied only as post-hoc temperature scaling.

233 4.3. Calibration methods and scenarios

234 We compare the no-recalibration baseline (None) against eight recalibration
235 methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-

ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and
 BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline
 plus these eight recalibration methods constitute the delivered set; CORAL, TTA,
 MC-Dropout, and Oracle are not delivered on real data (a registered deviation,
 see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply),
 the main analysis scenario (Sec. 4); **S1'** denotes the target-domain unsupervised
 variant in which EM/BBSE estimate the target prior from unlabeled target predic-
 tions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [26]):

$$\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $ECE_{\text{raw}}^{\text{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the tar-
 get corpus with raw softmax probabilities, and $ECE_{\text{TS}}^{\text{OOD}}$ is the same after source-
 cal-fit temperature scaling. Following the standard convention of Guo et al.
 [2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal:
 the predicted maximum probability against 0/1 top-label correctness; classwise-
 ECE is reported separately as a secondary endpoint. SmoothECE bandwidth:
 $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta ECE_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta ECE_{\text{OOD}} > 0 \quad (2)$$

The one-sided alternative $H_1 : \Delta ECE_{\text{OOD}} > 0$ follows the empirical asymmetry
 registered at revision A1: on ID-like data the source-fit temperature yields, empir-
 ically, little calibration change, whereas on a shifted target it is mismatched and a

257 non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a
 258 theorem: TS applied to an under-confident target can increase ECE, and our own
 259 data contain nine negative cells. The choice was made before the pilot results,
 260 which only triggered the endpoint re-location (revision A1). The one-sided test
 261 is the pre-registered choice, archived before multi-seed unblinding. Patient-level
 262 cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [15]; stratified pooling:
 263 strata = transfer pair $\times$ architecture.

264 The pre-registered CI method is BCa [15], at $B = 10,000$, re-run on the en-
 265 tire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The
 266 percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$;
 267 for the paired-cluster design with $B = 10,000$ and per-patient clusters, the dis-
 268 crepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were
 269 run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and
 270 magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain
 271 as historical provenance (5 lack log provenance, see Limitations); percentile CIs
 272 serve only as a sensitivity cross-check.

273 Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

274 The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion
 275 is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid*
 276 *re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered*
 277 *(percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

278 Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Or-
 279 acle recalibrator was implemented (`fit_oracle/apply_oracle`) but never ex-
 280 ecuted on real data, so G is undefined here and is excluded from all reported

281 results; see Limitations), $\text{decay} = \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descrip-
282 tive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept,
283 predicted/true prior L_1 .

284 Family structure: BH-FDR $q = 0.05$ over $6 \text{ pairs} \times 2 \text{ arch} \times 13 \text{ shift levels} =$
285 156 primary comparisons; the multi-seed validation is robustness evidence (60
286 seed experiments), not a confirmatory family.

287 4.5. Pre-registration revision A1 (transparency statement)

288 The primary endpoint was re-located from decay to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
289 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent
290 exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the
291 original ID-vs-OOD contrast. The revision direction is consistent with the empir-
292 ically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived
293 as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes.
294 The pilot results do not enter the main test report. All hypotheses in this paper are
295 tagged *exploratory + robustness validation*, not confirmatory.

296 The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times$
297 13 shift levels) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2$
298 $\text{architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining
299 13 shift levels designated as exploratory dose-response. Under the reduced family,
300 BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12
301 yields 2/12. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_`
302 `FAMILY_REDUCTION.md`.

303 4.6. *Pre-registration amendment A3 (corpus identity and MI provenance)*

304 A post-hoc audit of the CPSC corpus against the official challenge documen-
305 tation produced two wording corrections and one new disclosure, archived as
306 amendment A3 (docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md). The
307 registered protocol described the third corpus as “CPSC2018+2019”; it is in fact
308 **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-
309 Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task
310 with no diagnostic labels and never entered the corpus. The new disclosure is that
311 CPSC2018’s official nine-class taxonomy contains no MI, so the MI class of our 4-
312 class subspace is populated entirely by CPSC-Extra’s SNOMED annotations; un-
313 der a strict official-class-name caliber the subspace would reduce to three classes.
314 A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded
315 value, no split, and no analysis decision. The registered protocol snapshots are left
316 byte-identical and A3 is archived separately, so the before/after pair required by
317 pre-registration practice is preserved. The audit also found that 33 CPSC Database
318 records (A6845–A6877) were silently lost when the source archive was extracted
319 (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

320 4.7. *Decomposition of the benefit decay*

321 We decompose the retained benefit into slope / intercept / prevalence contribu-
322 tions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

323 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the
324 chain residual (zero under nested counterfactuals). Identifiability is checked via

325 the Fisher information determinant on a pre-registered threshold τ ; entangled re-
326 gions are reported on the error map without a threshold. Shapley attribution is
327 reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$)
328 mark uninterpretable shares.

329 4.8. Two-layer joint bootstrap

330 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$
331 joint Δ ECE CI (two-layer; honest non-nested approximation), per protocol §7.

332 4.9. Net Clinical Value (NCV)

333 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
334 the clinical decision impact of TS. For each (sample, class) pair we compare
335 the binary decision (threshold $\tau = 0.5$) before and after calibration, counting
336 TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-
337 calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} -$
338 $\text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s
339 patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-
340 FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves
341 clinical decisions. It is computed on ID and OOD splits (ncv_id, ncv_ood; E3
342 CSV).

343 4.10. Tooling

344 Parts of the analysis and plotting code were drafted and refactored with the
345 assistance of a large language model assistant (GLM), used as a supervised coding
346 assistant. All experiments and statistical analyses were executed with the author’s
347 own scripts, and every reported number is reproducible from the public repository

(Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

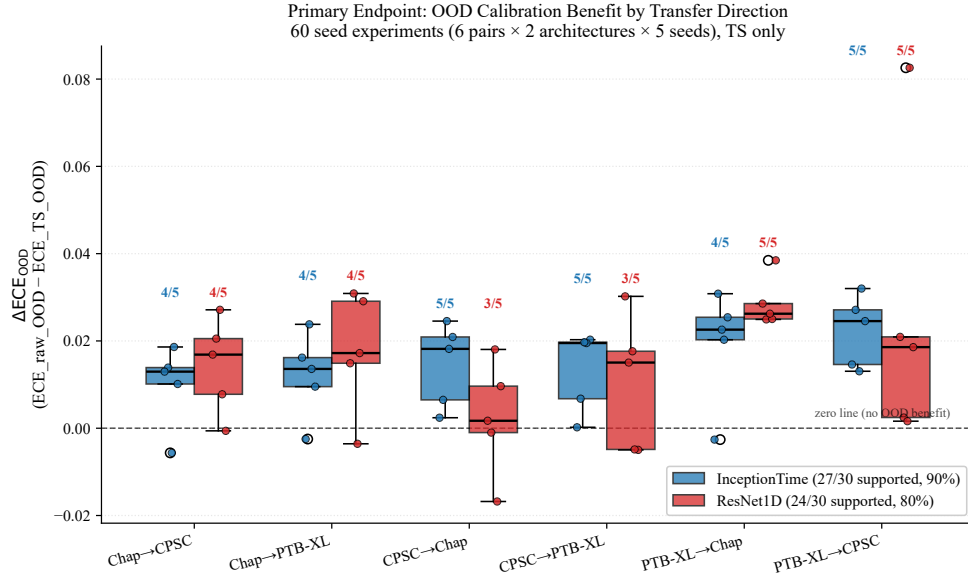


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap $\rightarrow$ CPSC, CPSC $\rightarrow$ Chap, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC) and $K = 5$ for the two CPSC-free pairs (Chap $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chap); see §4.1. “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC ($K=4$)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL ($K=5$)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
	RN	42	-0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	-0.0010	-0.0036	-0.0036	✗
		46	-0.0042	+0.0149	+0.0139	✓
CPSC→Chap ($K=4$)	IT	42	+0.0029	+0.0065	+0.0062	✓
		43	-0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	RN	42	-0.0002	-0.0010	-0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	-0.0152	-0.0168	-0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
CPSC→PTB-XL ($K=4$)	IT	42	+0.0029	+0.0068	+0.0066	✓
		43	-0.0012	+0.0203	+0.0198	✓
		44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
	RN	42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	-0.0050	-0.0051	✗
		44	-0.0032	-0.0049	-0.0049	✗
		45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap ($K=5$)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC ($K=4$)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

356 Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
357 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10,
358 zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The
359 nine counter-examples are explained in Sec. 6.1.

360 The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no
361 partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

385 qualified, the OOD benefit being about $1.9\times$ the ID benefit rather than an exclu-
386 sively OOD effect. The positive ID estimates reflect finite-sample calibration-split
387 variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model
388 selection on the endpoint family), all disclosed.

389 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

390 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on Inception-
391 Time. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

392 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most
 393 fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood
 394 mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60
 395 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed
 396 coverage on all 6 directions (Figure 4).

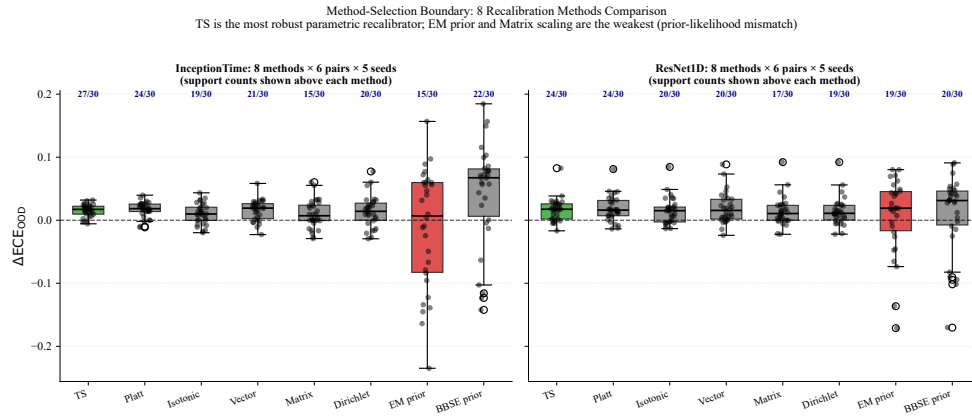


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

397 Across the 30 TS experiments per architecture, the standard deviation of
 398 ΔECE_{OOD} is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with
 399 ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, in-
 400 cluding an unexcluded favorable outlier at seed 44, $\Delta ECE_{OOD} = +0.0826$). Sup-
 401 port deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a
 402 direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields

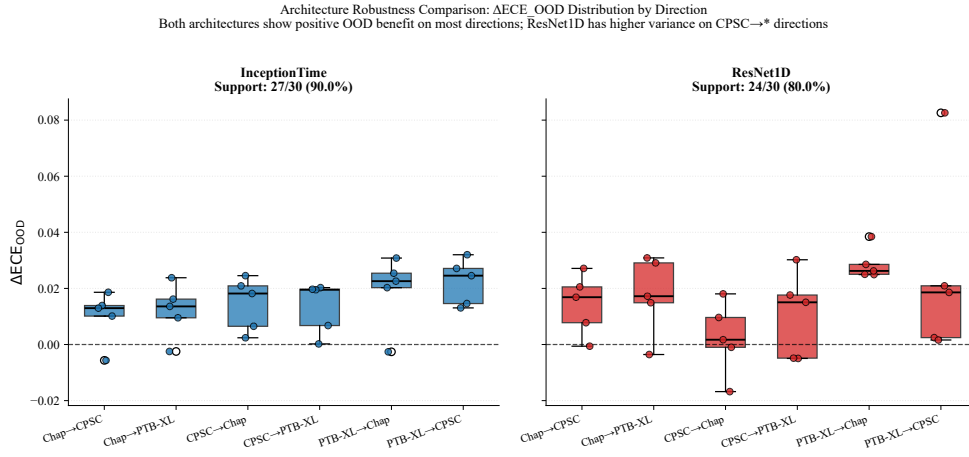


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat*: 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

403 $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%)
404 is not significant at $\alpha = 0.05$.

405 EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal,
406 then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated
407 confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies
408 rather than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
409 port on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-
410 signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

411 *Why TS and not Platt..* Platt achieves a marginally higher mean OOD benefit
412 than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table 3),
413 which invites the question of why TS is singled out as the recommended default.
414 Three measured criteria separate them, and all three favour TS. First, *robustness*
415 *of the sign*: TS supports 51/60 cells versus Platt’s 48/60, i.e. TS fails to certify
416 a positive benefit in three fewer experiments, which is the quantity a deployment
417 decision actually depends on. Second, *dispersion*: TS has a smaller across-cell
418 standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with
419 the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736,
420 EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

421 profile for a per-deployment recalibrator. Third, *the difference is not statistically*
 422 *distinguishable*: the patient-level paired contrast over the 60 matched cells gives
 423 $\overline{\Delta ECE}_{\text{TS}} - \overline{\Delta ECE}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with
 424 TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of
 425 Platt is therefore inside the noise, whereas the support advantage of TS is not a
 426 noise-level quantity: it is a count of cells in which the sign is certified. Given that
 427 the two methods are statistically indistinguishable on magnitude, the tie is broken
 428 on certification rate and dispersion, both of which favour TS. A further structural
 429 point reinforces the choice: TS is a single-parameter monotone transform and
 430 therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fit-
 431 ted independently per class, which generates each class’s score from a different
 432 monotone map and so does *not* preserve the ranking within a sample (the imple-
 433 mentation declares this explicitly in its docstring). TS is thus the only method
 434 in the delivered set whose calibration guarantee is obtained without perturbing
 435 the discrimination it leaves untouched (§5.9). None of this makes Platt wrong; it
 436 makes Platt a reasonable alternative whose mean advantage is not resolvable at
 437 $n = 60$, and whose certification rate is lower.

438 5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

439 The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in$
 440 $\{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposi-
 441 tion recovers ground truth within the pre-registered identifiable regime (Fisher
 442 information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the
 443 recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of
 444 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in ex-
 445 treme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most

446 clinically significant (see `decomposition_validation.csv`; the $n = 20,000$
 447 CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three
 448 entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times$
 449 $10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with
 450 the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-
 451 reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

452 The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE;
 453 its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of
 454 decomposition explanatory power) remains to be run on real data; per the protocol
 455 self-binding clause, “mechanism” wording is limited to “mechanism decomposi-
 456 tion (synthetic validation part)” until cross-fitting completes.

457 5.5. Predictability experiment (*failure branch triggered*)

458 The predictability experiment (protocol §8.5) tests whether an unsupervised
 459 target-domain statistic (predicted-prior L_1 distance, logit moments) predicts ben-
 460 efit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The imple-
 461 mented fold unit is (architecture, transfer pair) (12 folds), versus the registered
 462 pair-only folds (6); under that registered deviation the pair-only recomputation re-
 463 tains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime +
 464 ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

465 Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$.
 466 The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , trig-
 467 gering the **failure branch**: the three-step chain (decomposition + predictability +
 468 deployment) is downgraded to a **two-step chain** (decomposition + deployment),
 469 and the deployment criterion is renamed “empirical lookup table” rather than “pre-
 470 dictive criterion”. No post-hoc predictability claim is made.

471 Re-running the LOO R^2 with three label-free features (fitted temperature T ,
 472 Wasserstein-1 distance between source and target logit distributions, class-wise
 473 entropy difference) gives a sharper and less favourable picture for the bound-
 474 ary. Only T carries signal: under the implemented (architecture, pair) folds it
 475 reaches LOO $R^2 = +0.340$ (bootstrap CI $[+0.044, +0.363]$), and under per-cell
 476 folds $+0.361$ (CI $[+0.201, +0.584]$), while the Wasserstein distance (-0.064) and
 477 the entropy difference (-0.033) carry none. T is also a perfect *directional* predic-
 478 tor: $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells, against 51/60 (85%) for the
 479 constant-positive baseline. We therefore do *not* claim that no label-free quantity
 480 carries information about the endpoint — T does. The claim we retain is the
 481 pre-registered one: no label-free feature reaches the LOO R^2 threshold of 0.5
 482 under the implemented fold unit ($0.363 < 0.5$), so the failure branch stands and
 483 the three-step chain remains downgraded to two steps. Two qualifications are
 484 required. First, the verdict is fold-unit-sensitive: per-cell folds give a CI upper
 485 bound of 0.584, which would *not* trigger the branch, so the downgrade should be
 486 read as a property of the coarser (architecture, pair) unit as much as of the data.
 487 Second, T ’s predictive power is not independent evidence of a substantive bene-
 488 fit: $T = 1$ is a fixed point of the transform, so part of the relation is mechanical,
 489 and the sign rule is the direction temperature scaling must have by construction
 490 (Sec. 5.8). The gate in §5.9, based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination
 491 criterion, remains the appropriate actionable criterion.

492 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

493 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
 494 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
 495 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	−0.096	−0.720	+0.104	Failure branch
Platt	−0.059	−0.584	+0.002	Failure branch
Isotonic	+0.011	−0.569	+0.101	Failure branch
Vector	−0.061	−0.522	+0.073	Failure branch
Matrix	−0.137	−0.516	−0.013	Failure branch
Dirichlet	−0.084	−0.394	+0.032	Failure branch
EM prior	+0.143	−0.250	+0.274	Failure branch
BBSE prior	+0.136	−0.354	+0.201	Failure branch

496 SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , primary-endpoint conven-
497 tion). Comparators: always-none, always-TS, always-buy- n -labels.

498 Sensitivity / specificity (Table 6): Youden’s J is maximized at raw ECE
499 $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is
500 high (92% of beneficial cases caught) but specificity low (20% of non-beneficial
501 cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure:
502 missing a beneficial recalibration costs more than an unnecessary one. *Caveat*: the
503 threshold is searched on the same 1230-cell pool used for reporting (in-sample).
504 A leave-shift-type-out recomputation (threshold selected excluding the held-out
505 dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead
506 drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127
507 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE > threshold → predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

508 four held-out dose levels (results/deployment_holdout_recompute.csv).
 509 The in-sample numbers are therefore optimistic, and the criterion requires inde-
 510 pendent validation before operational use.

511 Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap,
 512 per-cell labels), requiring target ground-truth labels unavailable in the unlabeled
 513 stream. The only label-free proxies examined are the predictability experiment’s
 514 target statistics (predicted-prior L_1 distance, logit moments) plus the source-fitted
 515 temperature T ; that experiment triggered its pre-registered failure branch (LOO
 516 R^2 CI upper bound $0.104 < 0.5$, Table 5). T does carry signal (LOO $R^2 = +0.34$;
 517 sign of the endpoint correct in 60/60 cells), but it is a source-side quantity describ-
 518 ing the transform being evaluated rather than an independent target-side statistic
 519 (Sec. 5.8), so no label-free deployable criterion can be claimed. The lookup table
 520 is an *analysis- and audit-time decision aid* for a model owner with labeled eval-
 521 uation data; the actionable deployment path is small-sample recalibration: buy n
 522 labeled target records and recalibrate locally (protocol §5 S2). A deployable gate
 523 remains future work (Limitations xiii).

524 Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell
 525 satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement
 526 > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-
 527 based criterion was not evaluated), a safety rate of 0.0064. The source of
 528 this figure is `results/deployment_metrics.csv` (written 2026-09-05, i.e. be-
 529 fore the label-encoding incident described in the coverage note below; pro-
 530 duced by `scripts/step3_deployment.py` under the criterion $\Delta\text{ECE} < -0.01 \wedge$
 531 $\text{cal ECE} < 0.05$). It is *not* reproducible from the currently released L2 artifacts,
 532 because the per-run `l2_shift_results.json` files that feed it were overwritten
 533 on 2026-09-10 by outputs computed under the wrong label encoding (see the cov-
 534 erage note below); re-deriving the same criterion from those on-disk L2 outputs
 535 gives $6/156 = 0.038$ rather than $1/157$. The denominator deserves care, because
 536 157 is prime and therefore not a clean product of the grid dimensions. The stated
 537 $6 \times 2 \times 13 = 156$ is the BH-FDR *family* size ($6 \text{ pairs} \times 2 \text{ architectures} \times 13 \text{ lev-}$
 538 els); the 157 cells are the *deployment* snapshot, which covered $6 \text{ pairs} \times 2 \text{ seeds}$
 539 at 12 of the 13 shift levels ($6 \times 2 \times 12 = 144$) plus 13 (pair, seed) combinations at
 540 `fs250`: $144 + 13 = 157$. One (pair, seed) combination therefore appears at `fs250`
 541 only — a ragged-grid artefact of the two-seed snapshot, not a contradiction with
 542 the family size. The rate is also not method-specific: $1/157$ is identical for `ts`,
 543 `platt`, `vector` and `bbse_prior`, so the operating criterion is met at the same rare
 544 rate by every recalibrator in the grid. The claim that the safety criterion is rarely
 545 met is unaffected in direction (all values are small), but the exact rate must be
 546 regenerated. Sign convention: $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (nega-
 547 tive means improvement), opposite to the primary-endpoint convention. *Coverage*
 548 *note*: the L2 shift matrix has been completed on all 390 planned (pair, seed, shift)

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	−0.0145	0.0099	157
always-buy- n -labels	−0.2905	0.1861	1230
lookup-table	−0.0160	0.0289	155

549 combinations (6 pairs $\times$ 5 seeds $\times$ 13 levels; results/l2_shift_full_390cells.csv);
550 the safety figures quoted here are from the initial two-seed InceptionTime subset
551 (seeds 42–43, ResNet1D L2 excluded), so these numbers rest on 2-seed, single-
552 architecture coverage. The criterion is rarely satisfied, so TS is a useful default but
553 not a universally safe recalibrator, and should be combined with local validation.

554 At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 ver-
555 sus 0.0999 for treating all cells; by the 90th percentile it turns negative
556 (−0.0024). Net benefit over treat-all is modest, confined to low thresholds
557 (results/deployment_metrics.csv).

558 Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE
559 −0.0160) outperforms always-none (0.000) and matches always-TS (−0.0145)
560 on the beneficial subset, but is dominated by always-buy- n -labels (−0.2905), the
561 target-label upper bound. The lookup table needs no target labels and captures
562 most of the TS benefit.

563 5.7. Noise floor control: SmoothECE differential under perfect calibration

564 A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact
 565 of SmoothECE’s sensitivity to temperature scaling T under perfect calibration
 566 ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated pre-
 567 dictions ($T_{\text{true}} = 1$) we apply T and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) -$
 568 $\text{SmoothECE}(\text{raw})$.

569 The floor is strongly *distribution-dependent*: for flat or wide confidence
 570 distributions it is essentially zero, or slightly negative at moderate T . We
 571 therefore do not evaluate it on synthetic distributions. Instead we use the
 572 empirical top-label confidence distribution of each of the 60 main-grid cells
 573 (checkpoints/e2_probs_cache/*.npz), relabel each cell as perfectly cali-
 574 brated by resampling $y \sim \text{Bernoulli}(c)$ at its own confidence c (three draws per
 575 cell), and report the band across cells. This removes any freedom to choose a
 576 distribution that suits the argument, and it evaluates the floor where the study’s
 577 own predictions actually live (pooled median confidence 0.79, with 30% of sam-
 578 ples above 0.9). The synthetic three-distribution sweep (uniform, normal-skewed,
 579 bimodal at $n = 2,000$) is retained as a sensitivity analysis.

580 The noise floor grows monotonically in $|T - 1|$ and is distribution-dependent.
 581 Evaluated on the 60 main-grid cells — median across cells, with the cross-cell
 582 range in parentheses — $T = 1.0$ gives exactly 0 (exact cancellation); $T = 1.25$
 583 gives +0.0138 (−0.0115 to +0.0294); $T = 1.5$ gives +0.0389 (−0.0047 to
 584 +0.0597); $T = 3.0$ gives +0.1260 (+0.0404 to +0.1605); and $T = 4.5$ gives
 585 +0.1582 (+0.0605 to +0.2129). The synthetic sweep agrees in magnitude for
 586 high-confidence distributions ($T = 1.5$: +0.039 to +0.055) but not in sign for flat
 587 ones ($T = 1.25$: −0.009 for uniform), which is why the empirical distributions

are used. The floor is positive in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90% ($T = 1.25$) to 97% ($T = 1.5$) of cells in between. Observed raw ECE across the 62 raw-stage cells of the TS-component ablation under the corrected label encoding (results/ablation_ts_components.OLD_ENCODING.csv, stage=raw) ranges 0.036 to 0.357 (mean 0.211), i.e. $2\text{--}21\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show raw ECE well above the floor (0.065–0.321 vs. ≈ 0.017), so they are not perfect-calibration cases. At their own fitted temperatures ($T \in [0.93, 1.00]$) the floor is ≤ 0.013 , comparable to their $|\Delta\text{ECE}|$ (≤ 0.017), but its direction is opposite to theirs, so they are not floor artifacts; their common mechanism is given in Sec. 5.8. The direction of the floor is what matters for the endpoint, and it is fixed. In the primary convention ($\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{post}}$) the floor inflates ECE_{post} and therefore contributes *negatively*. The identity is the minimiser of ECE under perfect calibration, so the floor is positive in the noise convention on both sides of $T = 1$: in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90–97% of cells for $1.1 \leq T \leq 1.5$. The idealised endpoint is therefore $-\text{floor}(T) \leq 0$ almost everywhere, and a predominantly attenuating artefact cannot generate the positive endpoint observed in 51/60 experiments. We test this and three further implications of the floor hypothesis in Sec. 5.8.

5.8. Noise-floor mechanism test

The control above quantifies the floor; it does not by itself decide whether the floor *drives* the endpoint. We therefore test four implications of the “floor artefact” hypothesis, using the source-fitted temperature T recovered per cell from the TS-component ablation under the corrected label encoding.

612 (i) *Direction..* The floor contributes negatively to $\Delta\text{ECE}_{\text{OOD}}$ in 90–100% of cells
 613 at every $T \neq 1$ (Sec. 5.7), so the hypothesis predicts a systematically *negative*
 614 endpoint. The observed mean is +0.0159, positive in 51/60 cells.

615 (ii) *Sign pattern with T ..* In the corrected data $T \in [0.932, 1.498]$ (median 1.076;
 616 no cell reaches $T \geq 2$), and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ in 60/60 cells: all 51
 617 cells with $T \geq 1$ are positive (mean +0.0195) and all 9 cells with $T < 1$ are neg-
 618 ative (mean -0.0047). This is the direction temperature scaling is expected to
 619 have — $T > 1$ softens an over-confident target and lowers its ECE, while $T < 1$
 620 sharpens it and raises ECE — and it supplies a unified mechanism for the nine
 621 counter-examples of Sec. 6.1: they are exactly the nine cells with $T < 1$, in-
 622 cluding the extreme CPSC→Chapman / ResNet1D / seed 44 case ($T = 0.932$,
 623 $\Delta\text{ECE}_{\text{OOD}} = -0.0168$), the sharpest sharpening in the grid.

624 We are explicit that this is an *empirical* regularity over the range the fits oc-
 625 cupy, not a mathematical identity: ΔECE is computed from the same probabilities
 626 that define T , but the two are not algebraically linked, and the correspondence
 627 is falsifiable. Pushing a fixed temperature well past the fitted range breaks it —
 628 applying $T = 2.0$ to the same 60 cells flips 5 of them negative — and substituting
 629 the (mis-encoded) temperatures of the coverage note below drops the agreement
 630 from 60/60 to 56/62. Its role here is therefore *explanatory* (it accounts for the sign
 631 of every cell) rather than *exclusionary*; the exclusion of the floor hypothesis rests
 632 on (i), (iii), and the mechanism test below.

633 (iii) *Magnitude strata..* Under an additive floor the endpoint should be *smaller*
 634 where the floor is larger. $\text{floor}_{\text{hi}}(T)$ reaches the mean effect only for $T \geq 1.25$ (4/60
 635 cells); those cells give +0.0442 (4/4 positive) against +0.0138 (47/56 positive)
 636 elsewhere. Restricting to the $T \geq 1$ branch — 51 cells, all positive, so the sign

637 flip of (ii) cannot confound the comparison — and splitting at the median floor
638 (0.0078) gives +0.0117 (26/26) below and +0.0276 (25/25) above: the endpoint
639 is *larger* where the floor is larger, the opposite of an additive artefact. Since
640 the floor attenuates, these values are lower bounds. This stratum comparison is
641 nevertheless weaker evidence than (i)–(ii), because floor_{hi} is monotone in T on
642 this side while the endpoint’s growth with T is partly mechanical.

643 (iv) *Negative control and additivity.* The independent ablation channel (same
644 SmoothECE family) changes by -0.0124 on average and changes sign across
645 cells (-0.083 to $+0.093$), so the metric has no systematic positive push. The floor
646 at the cells’ own temperatures spans 0.0001 to 0.0595 , up to $54\times$ the median per-
647 cell bootstrap CI width (0.0011); an additive bias of that size would swamp the
648 signal, yet subtracting the mid-band floor at each cell’s own temperature leaves
649 $+0.0140$, i.e. 88% of the original $+0.0159$. The floor is therefore not a cell-level
650 additive offset, and we do not subtract it per cell.

651 *Consistency check.* The endpoint also rises with T (OLS slope $+0.181$, $R^2 =$
652 0.951 over the 60 cells). We report this but do not rely on it: $T = 1$ is a fixed
653 point at which the endpoint vanishes by construction, so part of this relation is
654 mechanical. Its role here is that its sign is *opposite* to the floor’s contribution,
655 which is the direction test of (i)–(ii).

656 *Caveats.* The floor is computed on idealised perfectly-calibrated relabellings
657 of our own confidence distributions, so it is not a calibrated model of real ECE
658 noise; the extreme stratum in (iii) has only 4 cells. The conclusion is exclusionary,
659 not causal: the floor cannot account for the primary finding, but this does not by
660 itself establish a causal mechanism.

661 5.9. Discrimination-aware calibration gate (C5)

662 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for
 663 deployment: a model whose discrimination has collapsed under shift may be re-
 664 calibrable but clinically useless. We define a two-layer gate:

- 665 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 666 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-
 667 class accuracy.

668 Across 12 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures), 7/12
 669 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC,
 670 CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (Incep-
 671 tionTime only). The other 5/12 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-
 672 XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls
 673 below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The
 674 strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but
 675 $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded
 676 too far to be clinically actionable.

677 This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer
 678 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2
 679 rejects). It supplements C1 by also asking whether discrimination is sufficient for
 680 the recalibrated predictions to be clinically useful.

681 5.10. Real-data Shapley cross-fitting and predictability boundary

682 The synthetic Shapley validation (C3) confirms decomposition recovery in the
 683 identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx$
 684 $\hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit

all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error +0.29, median +0.29). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors are *not* localised to any particular pair: they spread over five source→target directions (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1), so no single pair accounts for more than two of them. What *does* separate them is effect size rather than shift size: the nine mis-signed cells have $|\Delta\text{ECE}_{\text{OOD}}|$ mean 0.0047 (max 0.0168), against 0.0195 (max 0.0826) for the 51 correctly signed cells, a factor of 4.1; restricting to $|\Delta\text{ECE}_{\text{OOD}}| \geq 0.02$, the model recovers the sign in 23/23 cells. The linear approximation thus degrades where the effect itself is small, not where the shift is largest. We record one counterweight for completeness: the decomposition is not entirely information-free as an *ordinal* signal. Its rank correlation with the true $\Delta\text{ECE}_{\text{OOD}}$ is $\rho = +0.31$ ($p = 0.015$), and the sign of ρ is stable across all 60 leave-one-out folds (minimum +0.28). That is weak, and far below the pre-registered $R^2 > 0.5$ bar, but it is reproducible — unlike the directional accuracy, which is exactly the constant-positive baseline.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is neither a valid *predictor* (*how much*) nor a valid *diagnostic* (*whether*). Its 51/60 direction accuracy coincides exactly with the constant-positive baseline (a model that always predicts “TS helps” scores 51/60 on this grid), so it carries no directional information beyond the prior that TS usually helps; the quantity

710 that does recover the direction is the source-fitted temperature T (Sec. 5.5). For
 711 clinical translation, the gate in §5.9, using only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the
 712 discrimination criterion, is the appropriate criterion, not the Shapley-predicted
 713 magnitude.

714 5.11. Net benefit decision curve summary

715 Across all 60 seed experiments, the net benefit gain of TS over raw predictions
 716 (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60
 717 (83.3%) of experiments, with mean gain $+0.0027$ and range $[-0.001, +0.017]$.
 718 The directions with the largest decision-cost reduction are CPSC $\rightarrow$ PTB-XL
 719 (mean DCR $+0.0625$; net clinical value $+0.0312$) and CPSC $\rightarrow$ Chapman (mean
 720 DCR $+0.0375$; NCV $+0.0188$), which also have the largest E3-subsample
 721 $\Delta\text{ECE}_{\text{OOD}}$ (mean $+0.137$ and $+0.136$ respectively, under the E3 subsampled
 722 binned-ECE protocol with $n_{\text{id}} = n_{\text{ood}} = 20$ patients). The direction with negative
 723 net benefit is Chapman $\rightarrow$ CPSC (mean DCR -0.0150 ; NCV -0.0075), which
 724 is correctly rejected by the discrimination gate (C5). We note that the decision-
 725 benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks
 726 at PTB-XL $\rightarrow$ Chapman and PTB-XL $\rightarrow$ CPSC, means $+0.0240$ and $+0.0238$ on
 727 the full SmoothECE grid): the two protocols differ in estimator, class set, and
 728 subsample, so we treat the E3 analysis as a directionally supportive but protocol-
 729 distinct check, not a conversion of the primary endpoint. What survives is
 730 the gate-level conclusion: where the discrimination gate permits deployment,
 731 the calibration benefit does not cost decision performance; where it is negative
 732 (Chapman $\rightarrow$ CPSC), no calibration benefit should be deployed at all.

733 *5.12. Net Clinical Value (NCV) results*

734 The pre-registered NCV (Sec. 4.9) is BH-FDR corrected ($q = 0.05$) across the
735 two secondary endpoints (DCR, NCV).

736 Result: the OOD NCV is $+0.008125$ with 95% CI $[+0.0014, +0.0148]$
737 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS
738 yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292
739 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly
740 worsens ID decisions, consistent with the ID boundary condition (TS compresses
741 confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV
742 Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring
743 the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3
744 CSV (`c1_dcr_ncv_multiclass.csv`).

745 *5.13. Reliability Diagrams*

746 Figure 5 shows the 6-direction averaged reliability curve, comparing model
747 confidence against empirical accuracy before and after temperature scaling. The
748 summary curve uses bin-count weighted averaging across all 6 transfer directions.

749 Figures 6–8 show the per-direction reliability diagrams for the representative
750 configuration (ResNet1D, seed 42). In each panel, the red squares show the raw
751 model calibration curve and the green circles show the post-TS calibration curve.
752 The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show
753 the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 re-
754 samples).

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

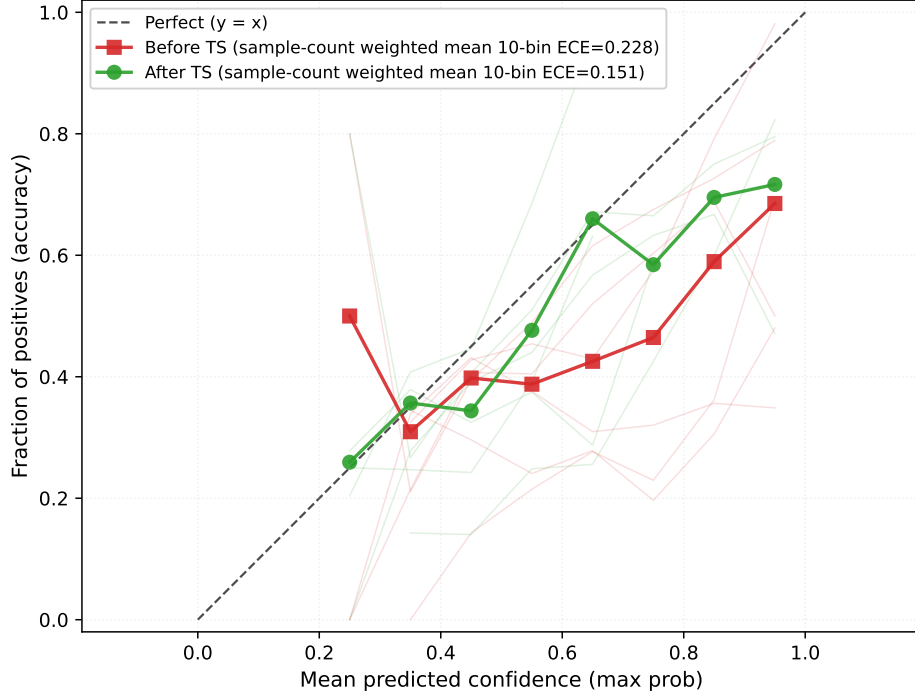


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently disclose all nine rather than excluding

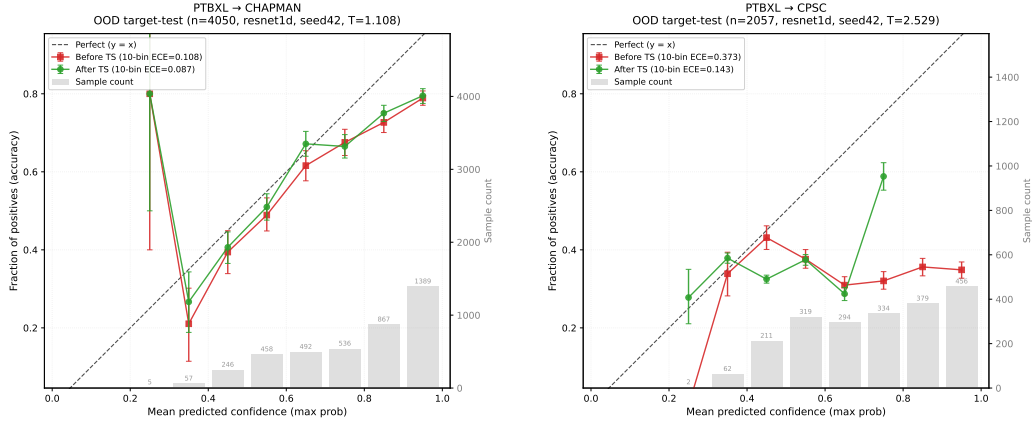


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

761 them, because the protocol requires all seed results to be reported and each has an
762 identifiable post-hoc mechanism supporting the qualified claim.

763 All nine share a single mechanism, stated in Sec. 5.8: they are exactly the nine
764 cells whose source-fitted temperature is $T < 1$. Because the temperature is fitted
765 on the source calibration split, $T < 1$ means the source model was *under*-confident
766 there; applying that temperature to the target sharpens an already over-confident
767 distribution and raises its ECE. The counter-example set is therefore identified by
768 a quantity available *before* the target is scored, and $\text{sign}(\Delta \text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$
769 holds in all 60 cells. The per-cell accounts below give the accompanying discrim-
770 inative and raw-ECE context; where a paragraph attributes the failure to “over-
771 correction”, that is the $T < 1$ regime.

772 Counter-example 1: Chapman→CPSC / InceptionTime / seed 45.
773 $\Delta \text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur:
774 lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44,
775 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature

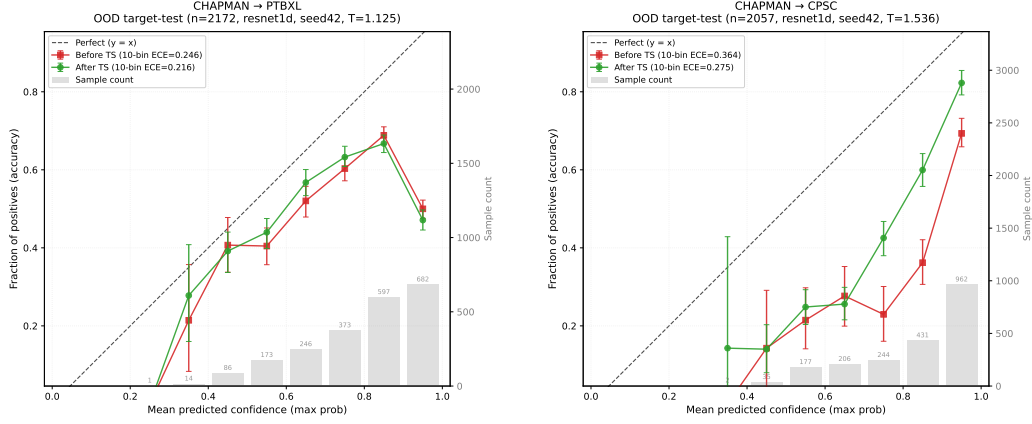


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

mismatch, not room shortage); and negative decay (-0.0075), the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD accuracy); the mechanism is seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

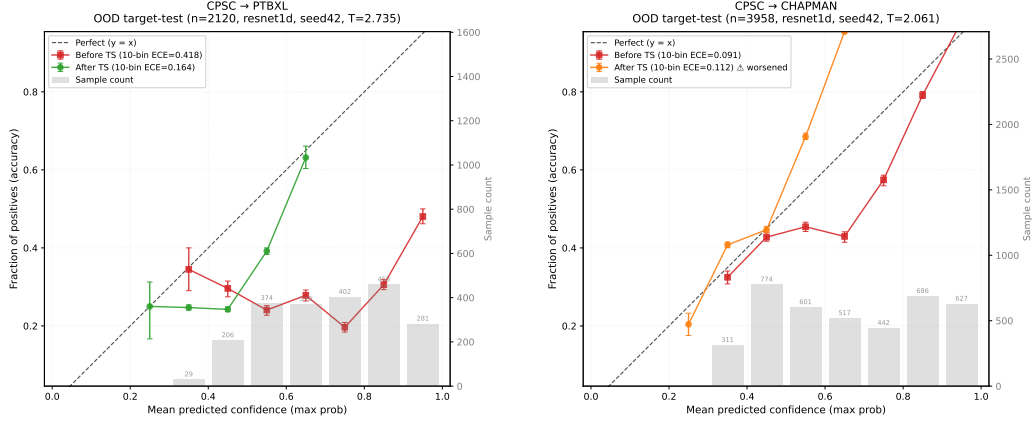


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the “low OOD accuracy + large gap” mechanism of the other 8.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

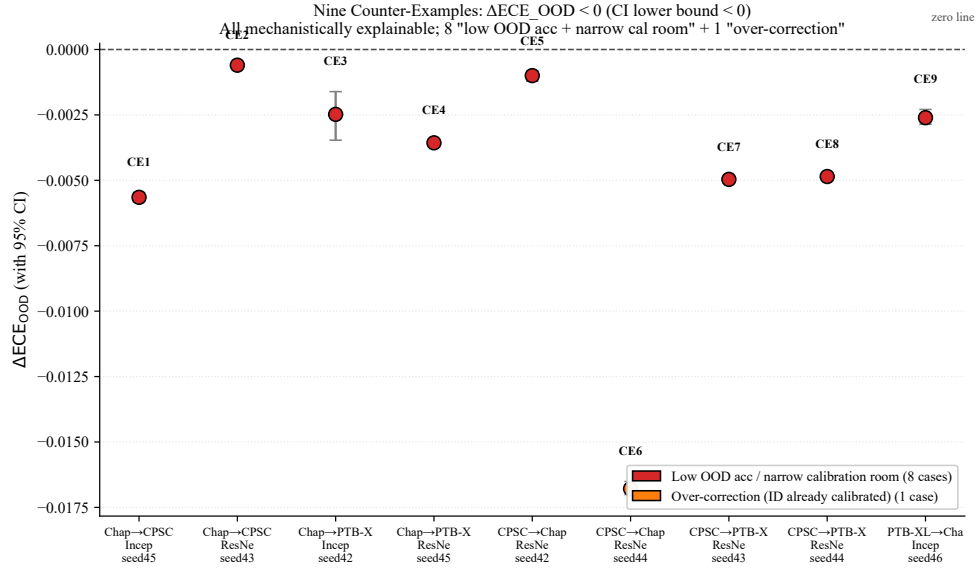


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

806 Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} =$
807 -0.0049 (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and
808 $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

809 Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46.
810 $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983
811 (lowest across the 5 seeds) and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, again
812 leaving nothing to correct.

813 Common pattern: eight of nine counter-examples co-occur with (a) low OOD
814 accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temper-
815 ature mismatch; the ninth (CE6) is over-correction on an already well-calibrated
816 backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so
817 the failure mechanism is seed-level temperature mis-transfer, not calibration-room
818 shortage. The counter-examples are local, explainable random variation; the other
819 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

820 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
821 ment. As a methodological boundary investigation, not a clinical deployment
822 guide, the relevant criterion is whether the effect is detectable and the mecha-
823 nism explainable, not a zero failure rate. All 9 counter-examples have identifiable
824 mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the
825 expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

826 6.2. *Prior correction vs. posterior correction robustness*

827 The method-selection boundary (Table 3) reveals a robustness asymmetry
828 between prior-correction methods (EM, BBSE) and posterior-correction meth-
829 ods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30
830 and 24/30 support respectively with small variance; prior-correction methods

831 achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449,
832 BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI
833 lower bound minimum -0.0763 , BBSE -0.0720). The mechanism is the prior-
834 likelihood mismatch: prior-correction methods fit a target prior that, when the
835 source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibra-
836 tion. Posterior-correction methods do not estimate a target prior and are therefore
837 immune to this failure mode, at the cost of lower mean benefit when the prior
838 correction *would* work.

839 6.3. Boundary condition: the ID benefit is small but nonzero

840 The ID boundary condition (C2) has a direct interpretation under the triggered
841 branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed
842 rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant
843 positive ID benefit (mean +0.0083 vs. OOD +0.0159; stratum means -0.0008 to
844 +0.0232, Table 2). The boundary condition does *not* mean TS is ineffective on ID
845 nor that the ID benefit is zero: the ID benefit is on average about half the OOD
846 benefit, so part of the apparent OOD gain is shared with the ID domain. A val-
847 ECE early-stopping effect (model selection within the endpoint family) plausibly
848 contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a
849 registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores
850 the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is
851 qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively
852 OOD effect. The ID-vs-OOD asymmetry is therefore a *difference in magnitude*
853 between two positive benefits, not presence versus absence.

854 6.4. Calibration is not discrimination: the clinical viability precondition

855 Temperature scaling is a monotone transform of the softmax probabilities;
856 it preserves the argmax by construction (asserted bitwise in the released sanity
857 check), so it cannot change classification accuracy, AUROC, or AUPRC. The pri-
858 mary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes
859 an already-deployable model *trustworthy*; it cannot make a non-discriminative
860 model deployable.

861 The accuracy statistics make this concrete. Across the 60 TS experiments,
862 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean
863 ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-
864 class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean
865 OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially
866 at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411
867 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620
868 (both well below 0.713). In those directions recalibration is moot, since no tem-
869 perature improves a model that should be replaced by a majority-class predictor,
870 consistent with the counter-example pattern (6/9 with low OOD accuracy). The
871 released artifacts include per-direction AUROC/AUPRC for all 60 experiments
872 (results/discrimination_metrics_60exp.OLD_ENCODING.csv, which now
873 carries both variant=raw and variant=ts rows): mean OOD AUROC ranges
874 from 0.735 (Chapman→PTB-XL) to 0.843 (PTB-XL→CPSC), i.e. far above the
875 accuracy-based majority-class reading yet direction-dependent; and TS changes
876 AUROC only marginally (mean $\Delta\text{AUROC}_{\text{TS-raw}} = +0.0004$, max |change| 0.002,
877 recomputed under the corrected label encoding and the corrected source-fitted
878 temperatures). We avoid the stronger phrasing that AUROC must be exactly

invariant: rescaling logits by $1/T$ is monotone *within* a row, so the argmax — and hence accuracy — is preserved exactly, but the cross-sample ranking of the top-label *score* is not, because the per-row normaliser differs across rows (measured Spearman between raw and TS top-label scores: 0.9945–0.9999). The discrimination-first gate—measure target discrimination before investing in recalibration—precedes the calibration decision rule in deployment.

6.5. *Explainable* $\neq$ *predictable*: post-hoc vs. pre-registered

We distinguish *explainable* from *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

The counter-examples are explainable, and they are also *locatable* in advance by one target-label-free statistic: all nine are the cells with $T < 1$ (Sec. 6.1). What fails is the stronger pre-registered requirement — a label-free feature that recovers the *magnitude* of benefit retention with LOO R^2 CI upper bound ≥ 0.5 — which is why the predictability experiment triggers the failure branch despite mechanistic explainability. Note that T 's sign rule is not a deployable criterion for deciding

902 whether to apply TS: it states the direction the transform must have by construc-
903 tion, not evidence that the transform helps (Sec. 5.8). The two-step chain (decom-
904 position + deployment lookup table) provides post-hoc explanation and actionable
905 guidance without a pre-registered predictability claim.

906 6.6. *Transparent re-location vs. HARKing*

907 The pre-registration revision A1 (primary endpoint re-location from decay to
908 $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [28, 27] is
909 whether the revision direction is theory-driven or result-driven. Our revision direc-
910 tion (“OOD is the calibration benefit source”) is consistent with the empirically
911 observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we
912 explicitly frame as an empirical prior rather than a theorem; the exploratory pilot
913 result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The
914 pilot results do not enter the main test report (protocol §7). All hypotheses in this
915 paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect
916 the post-revision status.

917 6.7. *Net clinical value: implications of the directional NCV asymmetry*

918 The NCV (Sec. 5.12) shows a directional asymmetry: OOD NCV =
919 $+0.008125$ ($p = 0.019$) is a small but significant positive value (recalibration im-
920 proves decisions on unseen target data more often than it degrades them); ID
921 NCV = -0.006292 ($p = 0.016$) is a small but significant negative value (a minor
922 decision-cost penalty on data the source model already calibrates well).

923 Two implications follow. (i) The **primary OOD benefit** (C1: 51/60 positive
924 on $\Delta\text{ECE}_{\text{OOD}}$) is supported: the positive OOD NCV shows the calibration gain

925 translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-**
 926 **cost improvement claim** (C5: 7/12 strata deployable) is partially qualified: the
 927 negative ID NCV tempers source-domain deployability. We therefore frame the
 928 NCV as *supportive of OOD deployment* and *cautionary for ID deployment*. It also
 929 informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to
 930 CI-width heterogeneity across splits. A permutation test with tighter CIs and a
 931 multi-threshold NCV analysis are registered as future work.

932 6.8. Limitations and future work

933 Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on all
 934 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba
 935 is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX
 936 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-
 937 step); the deployment criterion is an empirical lookup table, not a predictive crite-
 938 rion; (iii) real-data Shapley cross-fitting (§5.10) shows 85% direction accuracy but
 939 systematic magnitude overestimation (mean error $+0.29$), and a sensitivity anal-
 940 ysis with T , Wasserstein-1, and entropy features does *not* confirm a robust pre-
 941 dictability boundary: only T predicts ($\text{LOO } R^2 = +0.34$, and $\text{sign}(\Delta \text{ECE}_{\text{OOD}}) =$
 942 $\text{sign}(T - 1)$ in 60/60 cells), so the retained claim is the weaker one that no label-
 943 free feature reaches $R^2 = 0.5$ under the implemented fold unit (Sec. 5.5); (iv)
 944 the 27-cell decomposition validation uses binned ECE, not the main endpoint’s
 945 SmoothECE, so quantitative conclusions cannot be extrapolated across ECE def-
 946 initions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS
 947 safety rate 0.0064); TS is a useful default, not universally safe; (vi) nine counter-
 948 examples of 60 experiments (15.0%) are transparently disclosed, with a frequent
 949 post-hoc pattern (low OOD accuracy, non-positive decay, elevated OOD raw ECE)

950 but no post-hoc exclusion; (vii) the operating CI is BCa at $B = 10,000$, re-run on
 951 all 60 experiments (60/60 successful, metadata in `methods.ts.meta`); percentile
 952 CIs from $B = 200$ in 21/60 experiments are retained as historical provenance and
 953 are no longer used for any reported number. All 60 main-grid cells carry com-
 954 plete $B = 10,000$ /BCa provenance; the only cells lacking such metadata are the
 955 two excluded BiMamba toy runs (PTB-XL→Chapman, seeds 42–43), which do
 956 not enter the main endpoint. (viii) the Youden threshold is fitted in-sample (the
 957 protocol’s held-out dose levels were not used), making the operating character-
 958 istic optimistic; (ix) the L2 deployment/safety grid has since been completed on
 959 all 390 planned (pair, seed, shift) cells (`results/l2_shift_full_390cells.csv`), and
 960 the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise
 961 Stress Test Database (unavailable under PhysioNet licensing); (x) temperature
 962 was fitted once per experiment, so CIs omit temperature-fitting uncertainty (the
 963 pre-registered two-layer bootstrap is implemented but not enabled); (xi) model
 964 selection used validation ECE rather than the pre-registered NLL early-stopping
 965 rule, plausibly contributing to the positive ID boundary estimates; (xii) the per-
 966 pair decomposition into covariate, prior-probability, and concept-shift compo-
 967 nents is not asserted, since each pair mixes all three, isolated at eval time; (xiii)
 968 the deployment lookup table requires labeled target data, and the label-free prox-
 969 ies failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS
 970 is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC; the ar-
 971 tifacts now include per-direction AUROC/AUPRC for all 60 experiments (`re-`
 972 `sults/discrimination_metrics_60exp.OLD_ENCODING.csv`), so the viability pre-
 973 condition is screened by both accuracy and discrimination (OOD AUROC means
 974 range 0.735–0.843 by direction), with OOD accuracy below the majority-class

975 baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direc-
 976 tionally asymmetric ($\text{OOD} = +0.008125$, $p = 0.019$, positive; $\text{ID} = -0.006292$,
 977 $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting
 978 target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$
 979 to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits,
 980 with 48 as future work; (xvii) this work is from a single-author study, limiting inde-
 981 pendent review and audit, partially mitigated by the pre-registration protocol and
 982 public repository; (xviii) the secondary endpoint G (recoverable loss versus an or-
 983 acle recalibrator) is *not delivered*: `fit_oracle/apply_oracle` are implemented
 984 but were never executed on real data, so G is undefined in this study and no G
 985 result is reported anywhere in this paper or the repository; (xix) no signal-quality-
 986 index (SQI) stratification is performed: the target corpora are pipeline-processed
 987 releases and the L2 noise levels are synthetic, so the calibration benefit is not
 988 characterised as a function of measured signal quality (pSQI/kSQI-type indices);
 989 this is the principal gap for a signal-processing readership and is the first item of
 990 future work; (xx) no test–retest or repeatability analysis of the fitted temperature
 991 is possible within this data regime: all three public corpora are cross-sectional
 992 and provide a single recording per patient (Chapman-Shaoxing reports exactly
 993 one ECG per patient, §3), so same-patient across-session stability of T cannot be
 994 estimated from these data and is deferred to prospective multi-session collection.
 995 Future work: (1) prospective multi-device validation; (2) continuous shift spaces;
 996 (3) π -independent recovery estimators (BBSE/EM-class): BBSE is now delivered
 997 on the real grid (Honesty item 1), extension to further estimator classes remains;
 998 (4) the ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to
 999 test for a systematic interaction; (5) confirmatory testing under the A2 reduction

1000 (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6) re-
1001 running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU;
1002 (7) a multi-threshold permutation test on NCV.

1003 7. Conclusion

1004 We pre-registered and validated a multi-seed calibration boundary study of
1005 temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60
1006 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%,
1007 ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells
1008 (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile:
1009 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch;
1010 the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells
1011 pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60),
1012 EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both
1013 layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
1014 failures. The predictability failure branch is triggered (three-step chain becomes
1015 two-step), and all hypotheses are exploratory + robustness validation, not confir-
1016 matory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064),
1017 and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

1018 A noise floor control (§5.7) bounds, but does not eliminate, a SmoothECE ar-
1019 tifact: at moderate temperatures metric noise is $2\text{--}21\times$ below the observed raw
1020 ECE (0.036–0.357, mean 0.211), whereas for large $T > 3$ the floor (0.10–0.16) is
1021 comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be ex-
1022 cluded there. Real-data Shapley cross-fitting (§5.10) gives 85% direction accuracy
1023 with magnitude overestimation (mean error +0.29). The predictability boundary

1024 holds only in the weak sense that no label-free feature reaches $R^2 = 0.5$ under the
1025 implemented fold unit: the source-fitted temperature T does predict (+0.34, and
1026 the sign of $\Delta\text{ECE}_{\text{OOD}}$ in 60/60 cells), while Wasserstein-1 and entropy features do
1027 not.

1028 The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
1029 (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird ran-
1030 dom effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossAr-
1031 chitecture +0.0148). We caution, however, that the meta-analytic independence
1032 assumption is *not* satisfied here: the 60 cells share the same three corpora,
1033 patient sets, and architectures, and their between-cell Q statistics are extreme
1034 ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow.
1035 The interpretable pooled uncertainty is therefore the cluster-robust interval over
1036 the 12 pair $\times$ architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11;
1037 stratum SD 0.0072, about $1800\times$ the single-experiment bootstrap SE). The point
1038 estimate 0.0148–0.0159 and the count-based proportion ($51/60 = 85.0\%$) stand.

1039 The 60 experiments form the robustness-evidence family; amendment A2 re-
1040 duces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypothe-
1041 ses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$), with all 13 L2 shift levels
1042 re-designated as exploratory dose-response. BH-60 and Bonferroni leave the ro-
1043 bustness headline unchanged ($51/60$); the confirmatory family is BH-12 with 9/12
1044 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the
1045 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells
1046 fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells
1047 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60

1048 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE
1049 variance; they do not affect the sign-based support/counter-example classification.

1050 **Honesty Declaration (5 items, adversarial audit product)**

- 1051 1. MAPE_π is a **design check** (resampling constant read-back), not an esti-
1052 mator capability test. The independent π recovery registered as deferred
1053 (protocol §13, item 7) has since been delivered on the real 60-cell grid
1054 (`scripts/real_prior_recovery.py`): with target labels withheld, BBSE
1055 reduces the prior-recovery error from 0.1916 (source-prior baseline) to
1056 0.1132 — a 40.9% relative reduction, better than the baseline in 56/60
1057 cells, positive in 6/6 transfer pairs, with a pair-clustered bootstrap CI of
1058 $[+0.048, +0.111]$. EM gives no improvement (0.1923). Two limits remain
1059 and are reported as such: the absolute error is still 0.113, and the confusion-
1060 matrix condition number exceeds 10^3 in 5/60 cells and 10^4 in 1/60, so recov-
1061 ery is numerically boundary-sensitive there. **This does not repair the de-**
1062 **composition estimator**, which targets a different quantity (the target class
1063 prior, not the attribution coefficients) and whose real-data cross-fitting error
1064 remains +0.29 (Sec. 5.10).
- 1065 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
1066 **pretable as shares**; ranking uses absolute- value CIs.
- 1067 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the
1068 archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information
1069 mode), reported via exit code 2.
- 1070 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
1071 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement

1072 demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13).
1073 The wording is limited to “mechanism decomposition (synthetic validation
1074 part)”.
1075 5. The repository’s `scripts/preprocess_cinc2021.py`,
1076 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
1077 from the third-party SafeECGMatch release (original README in
1078 `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment
1079 grid.

1080 **Data availability**

1081 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018 (CPSC Database
1082 and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet un-
1083 der their respective licenses. The pre-registered protocol v2.1-A1, its hash
1084 manifest (`docs/osf_archive_manifest.json`), the revision A1 docu-
1085 ment, and the full per-experiment result files (`checkpoints/transfer/**/`
1086 `transfer_result.json`, `results/*.csv`) are included with the submission
1087 as supplementary material and are also available in the public repository
1088 <https://github.com/Toni-cs/ecg-calibration-transportability>.
1089 The protocol, both amendments (A1, A2), and a per-file archival-status re-
1090 port are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
1091 `docs/osf_archive_manifest.json` is the internal snapshot taken on 2026-09-
1092 05; it is not a substitute for the public record.

1093 **Code availability**

1094 The full pipeline code, preprocessing (reproducible from the public source
1095 databases), patient-level split generation, model training, the calibration-method
1096 implementations, and all statistical analyses, together with the environment
1097 lock file, the patient-level split indices for all five dataset variants and five
1098 random seeds, and selected per-experiment result artifacts, is publicly available
1099 at <https://github.com/Toni-cs/ecg-calibration-transportability>.
1100 Trained model checkpoints for the 60-experiment transfer grid are provided as
1101 release assets of the same repository. The repository will be archived on Zenodo
1102 with a DOI upon acceptance. No new raw waveforms were collected or generated;
1103 the patient-level split indices and label-mapping tables released with the code
1104 fully determine the data configuration used in every experiment.

1105 **Declaration of generative AI and AI-assisted technologies in the writing pro-** 1106 **cess**

1107 All scientific work reported in this study—the study design, the pre-registered
1108 protocol and its amendments, the implementation and execution of the 60-
1109 experiment transfer grid, the statistical analysis, and the interpretation of the
1110 results—was performed by the author. A large language model assistant (GLM)
1111 was used in two supervised roles, neither of which produced scientific content:
1112 as a coding assistant that drafted and refactored parts of the analysis and plotting
1113 code (Section 4.10), and as a language-editing aid that improved the wording and
1114 readability of text the author had written. The tool generated no experimental data,
1115 results, or conclusions, and made no analytical or interpretative decisions; every
1116 reported number is produced by the author’s scripts and is reproducible from the

1117 public repository. After using the tool, the author reviewed and edited all affected
1118 code and text as needed and takes full responsibility for the content of the pub-
1119 lished article.

1120 **CRedit authorship contribution statement**

1121 **Toni Guan:** Conceptualization, Methodology, Software, Validation, Formal
1122 analysis, Investigation, Data curation, Writing – original draft, Writing – review
1123 & editing, Visualization, Project administration.

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1131 **Declaration of competing interest**

1132 The author declares no competing financial or personal interests.

1133 The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5
1134 seeds; calibration curves, reliability diagrams, and Δ ECE bars per experiment)
1135 are provided as a separate supplementary file (supplementary.pdf, §1) and are
1136 additionally released with the code repository.

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